

PROBLEM STATEMENT

The Eight Puzzle Problem is a classic problem in Artificial Intelligence that consists of a **3 × 3 grid** containing eight numbered tiles and one empty space (blank). The objective is to rearrange the tiles from a given initial configuration to a specified goal configuration by sliding one tile at a time into the empty space. Only horizontal and vertical movements of adjacent tiles are allowed. The challenge is to find a valid sequence of moves that transforms the initial state into the goal state while satisfying the movement constraints.